

Gagan Kharoud(Bawa)

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Frontend Architect with **14+ years of experience** delivering **scalable, high-performance web applications** across energy, healthcare, and telecom industries. Expertise in **React, Redux, Next.js, Micro Frontends, PWAs, and AEM**, with a strong foundation in full-stack development.

Proven track record of **driving modernization initiatives, performance engineering, and frontend architecture** to deliver measurable business impact. Skilled at **integrating RESTful APIs, optimizing performance, building reusable component libraries, and ensuring accessibility (WCAG/ADA)**.

Recognized as a **technical leader and mentor**, leading cross-functional teams, defining frontend architecture standards, and aligning UI/UX with business goals. Thrive in **Agile/DevOps environments**, leveraging innovation and emerging technologies to continuously improve scalability, security, and user experience.

PROFESSIONAL EXPERIENCE

CONSTELLATION ENERGY

MD

Lead Frontend Engineer - Remote

April 2024 - Current

- Developed and maintained responsive, high-performance web applications using **React (Hooks, functional components), Next.js, TypeScript, and JavaScript (ES6+)**, ensuring scalability and performance across all modern browsers and devices.
- Helped team to integrate front-end components with Atlassian Jira APIs and ecosystem tools, enabling automated workflows, certification tracking, and enhanced collaboration within Agile teams.
- Collaborated with backend engineering teams to connect **UI components to Azure-hosted APIs and microservices**, optimizing API contracts, JSON payloads, and ensuring efficient data exchange for high-volume transactional processes.
- Implemented advanced state management solutions using **Redux Toolkit, React Query, and Context API**, simplifying complex asynchronous data flows and improving maintainability of enterprise applications.
- Applied **Azure DevOps CI/CD** pipelines to streamline builds, testing, and deployments into Azure Cloud (App Services and Static Web Apps), improving release velocity and reducing deployment risk.
- Utilized **Git and pull request workflows** to enforce collaborative development practices, maintain clean branching strategies, and ensure high-quality code reviews in a fast-paced environment.

- Developed and maintained automated unit and integration tests with **Jest, React Testing Library, and MSW**, increasing overall test coverage and accelerating QA sign-off.
- Monitored and optimized application performance with Azure Application Insights, proactively identifying issues, improving page load times, and maintaining system reliability.
- Led accessibility and responsive design initiatives, delivering **WCAG-compliant interfaces** and adaptive layouts that scaled seamlessly across devices and customer channels.
- Refactored legacy React modules into optimized **Next.js components**, implementing code splitting, lazy loading, batched rendering, and debouncing techniques to improve responsiveness and handle bulk workflows without degradation.
- Built reusable component libraries with **Material-UI**, establishing consistent design patterns and improving developer productivity across distributed teams.
- Worked with the team to design secure authentication/authorization flows leveraging OAuth 2.0 and Azure Active Directory (Azure AD), ensuring compliance with enterprise security standards.
- Actively participated in Agile ceremonies including sprint planning, backlog grooming, and daily stand-ups using Jira Agile boards, ensuring alignment with product owners and business stakeholders.
- Mentored engineers on React/Next.js best practices, Azure Cloud deployments, and CI/CD workflows, while contributing to technical documentation and coding standards that improved onboarding and long-term maintainability.

Key Achievements:

- Refactored legacy React modules into optimized Next.js components with **code splitting and lazy loading**, improving responsiveness.
- Built **reusable component libraries** with Material-UI, reducing feature delivery time by **30%**.

Key Technologies: React, Next.js, TypeScript, JavaScript (ES6+), Redux Toolkit, React Query, Context API, Material-UI, Atlassian Jira APIs, Azure App Services, Azure Static Web Apps, Azure Application Insights, Azure DevOps, Jest, React Testing Library, Git, OAuth 2.0, Azure Active Directory (AD), WCAG Accessibility

CIGNA (EXPRESS SCRIPTS - ESSDS)

Lead Frontend Advisor - Remote

New Jersey

Aug 2022 - Mar 2024

*As part of the architecture team, I was responsible for driving modernization initiatives and designing a **scalable, secure, and high-performance frontend platform**.*

- Architected and led migration of legacy systems to Next.js, leveraging React Server Components and App Routing to improve scalability, application security, and overall performance.

- Designed and implemented a Micro Frontend architecture using Module Federation, enabling independent deployments, stronger dependency isolation, and reduced security exposure across applications.
- Established a centralized Storybook-driven component library, enforcing design consistency, accessibility compliance (WCAG/ADA), and reusability across multiple projects.
- Drove frontend performance optimization initiatives, including code splitting, lazy loading, bundle size reduction, and asset compression, leading to measurable improvements in page load times and user experience.
- Partnered with backend teams to design secure, high-performing APIs, enforcing strict contracts, CORS policies, and optimized data flows that improved response times and reduced client-server overhead.
- Strengthened authentication and authorization architecture by implementing JWT-based auth and role-based access control (RBAC), ensuring secure, enterprise-grade access management.
- Introduced scalable state management strategies with Redux Toolkit, including API caching and optimized asynchronous data fetching to handle complex workflows efficiently.
- Enhanced overall code quality and maintainability through static analysis tools (ESLint, Prettier), strict architectural guidelines, and best practices for component structure and state management.
- Elevated test coverage and reliability by leading efforts in unit, integration, and end-to-end testing with Jest, React Testing Library, and Cypress, reducing regression risk in large-scale releases.
- Actively conducted architecture and security reviews, ensuring adherence to secure coding standards (XSS prevention, CSRF protection, input sanitization) and reducing vulnerability risks.
- Mentored engineering teams on performance-first React architecture, micro frontends, and secure coding practices, helping mature frontend development standards across the organization.

Tech Stack:

Frontend Technologies:

- React.js, Next.js, Javascript, Typescript,, Redux Toolkit, React Hook Form, Storybook, React Tables (for data visualization and interactive tables), agGrid, Axios

Backend & Dev Ops:

- Node.js, Express.js, BFF architecture, RESTful APIs, Redis, MongoDB
- Azur, Jenkins (for CI/CD automation)

Software Design & Best Practices:

- Micro Frontends (using Module Federation), SOLID Principles, DRY, KISS, YAGNI

VERIZON WIRELESS

Lead Frontend Engineer - Remote

Irvine, CA

March 2015 - July 2022

- Architected modernization initiatives, migrating large-scale consumer portals from JSP/Backbone to React/Redux, and later to Next.js with Micro Frontends and SSR, enabling independent deployments and improved SEO.
- Drove performance engineering, achieving 50% faster load times and 25% higher traffic/conversions through caching strategies, bundle optimization, and A/B testing.
- Built reusable component libraries and enforced accessibility standards (WCAG/ADA), reducing feature delivery time by 30%.
- Implemented AEM-integrated React SPAs, enabling business teams to manage content dynamically and accelerate time-to-market.
- Collaborated on backend workflows with Play, Akka, Redis, and Node.js to scale real-time, high-volume systems.

Migration to React/ Redux / Play/ Akka/ Redis

- Participated in the migration of a legacy application from Backbone JS to a modern, message-driven, reactive application built with Play, Akka, Redis, Node, React, and Redux..
- Developed and implemented workflows and guidelines to ensure code consistency, readability, maintainability, and testability. Assisted in the implementation of server-side rendering using Node and React to improve search engine optimization (SEO) and enhance performance
- Participated in the migration of a legacy application from JSP to Backbone JS, playing a key role in modernizing and upgrading the application's technology stack.

VERIZON WIRELESS **Frontend Developer**

Irvine, CA
March 2013 - Feb 2015

Migration to ATG / jQuery/ jQuery UI/ jQuery Mobile

- Participated in the migration of a legacy application to a new JSP/DSP - ATG platform, and also assisted in the transition from an ATG monolithic architecture to a headless architecture using Backbone JS, Handlebars/Underscore, and microservices-based technologies. Contributed to the modernization and improvement of the application's technology stack and design.
- Implemented Sitecatalyst tagging and Ensign tags on all pages to enable comprehensive reporting and analytics. Contributed to the implementation of the Grunt task runner to streamline and automate various development tasks.

Mitsubishi Electric Automation, Inc.(MEAU) **UI Developer**

Vernon Hills, IL
May 2012 - Dec 2012

- Involved in revamping web pages using jQuery, jQuery Mobile and jQuery UI
- Fixing defects and completing Change Requests and Update Daily status

- Involved in Engenio Data migration project, responsible for creating and reviewing Test Cases, Update RTM
 - Enhancements Requests from Business for data mapping and Experience Gaps.
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TECHNICAL SKILLS

Frontend Development:

- JavaScript & Frameworks: React.js, Redux, Redux Toolkit, Next.js, Node.js, Express.js, GraphQL, WebView
- Legacy & Alternative Libraries: Backbone.js, jQuery, jQueryUI, Mootools, Handlebars.js, Underscore.js, Hogan.js, Mustache.js
- Component Development: Storybook (UI Component Library)

Styling & UI Frameworks:

- CSS & Preprocessors: Tailwind CSS, Styled Components, SCSS/SASS, LESS
- CSS Methodologies: BEM, SMACSS
- UI Frameworks: Bootstrap, Materialize, AEM (Adobe Experience Manager)

Performance, Security & Optimization:

- Performance Tuning: Lighthouse, Server-Side Rendering (SSR), Lazy Loading, Code Splitting
- Security & Fraud Prevention: Frontend Security Best Practices, Authentication/Authorization, CORS, CSRF Protection

Build Tools & DevOps:

- Tooling & Automation: Babel, NPM, Grunt
- CI/CD & Deployment: Webpack, Jenkins, Azure DevOps, GIT, Agile/Scrum
- Microservices & Server-Side Integration: ATG Monolithic, Microservices Architecture

Monitoring & Analytics:

- Logging & Monitoring: Kibana, Grafana
- User Analytics & Behavior Tracking: Medallia, Content Square, BrowserStack

Prototyping & UI/UX Collaboration:

- Design & Wireframing Tools: Figma, InVision

EDUCATION

Masters in Information Systems - Central Michigan Univ, MI, USA (2011)

Bachelors of Computer Applications - Punjab Technical Univ, India (2007)

Post Graduate Diploma in Computer Applications - Punjab Technical Univ, India(2008)